Table 1: Exploratory three-model MAE ($\times 10^{-3}$): five-seed mean $\pm$ sample SD. The independent test geometry count is n.

Protocol	n	Matched set	Sum GNN	Set attention
IID	160	8.503 ± 0.107	8.699 ± 0.398	9.174 ± 0.227
LCRO (9–15)	264	11.785 ± 0.615	11.139 ± 0.352	11.776 ± 0.480
ISO 834	158	8.395 ± 0.333	8.368 ± 0.374	8.683 ± 0.377
ASTM E119	95	8.176 ± 0.198	7.839 ± 0.337	8.223 ± 0.392
Perturbed ISO	111	8.807 ± 0.071	8.725 ± 0.205	9.209 ± 0.501
Plateau	103	9.796 ± 0.492	9.328 ± 0.337	10.970 ± 0.412
Decay	113	13.858 ± 0.818	14.826 ± 2.245	13.046 ± 1.167
Log variant	69	13.770 ± 0.431	13.829 ± 0.926	14.632 ± 1.211
External fire	49	10.526 ± 0.593	10.253 ± 0.154	10.881 ± 0.965
Bilinear	77	31.603 ± 5.587	31.003 ± 5.287	27.890 ± 1.632
Linear	125	26.293 ± 7.775	25.338 ± 6.719	34.675 ± 12.092
Smoldering	97	12.105 ± 3.371	11.603 ± 1.341	12.517 ± 2.467

Table 2: Exploratory paired MAE differences ($\times 10^{-3}$) with 95% crossed geometry–seed bootstrap intervals. Positive values favor attention.

Protocol	n	Matched set − attention	Sum GNN − attention
IID	160	−0.671 [−1.112, −0.275]	−0.475 [−0.942, +0.002]
LCRO (9–15)	264	+0.009 [−0.624, +0.651]	−0.637 [−1.215, −0.008]
ISO 834	158	−0.288 [−0.723, +0.169]	−0.315 [−0.743, +0.106]
ASTM E119	95	−0.047 [−0.580, +0.435]	−0.384 [−0.808, +0.048]
Perturbed ISO	111	−0.401 [−0.940, +0.089]	−0.484 [−1.155, +0.132]
Plateau	103	−1.174 [−1.835, −0.516]	−1.643 [−2.375, −1.025]
Decay	113	+0.812 [+0.085, +1.525]	+1.780 [+0.386, +3.410]
Log variant	69	−0.862 [−2.145, +0.346]	−0.802 [−2.080, +0.458]
External fire	49	−0.354 [−1.820, +0.925]	−0.628 [−1.719, +0.399]
Bilinear	77	+3.713 [−0.586, +8.993]	+3.114 [−2.298, +8.764]
Linear	125	−8.383 [−20.802, +0.389]	−9.338 [−21.435, +1.896]
Smoldering	97	−0.411 [−1.479, +0.523]	−0.913 [−3.352, +1.433]